

Complements of Operational Research

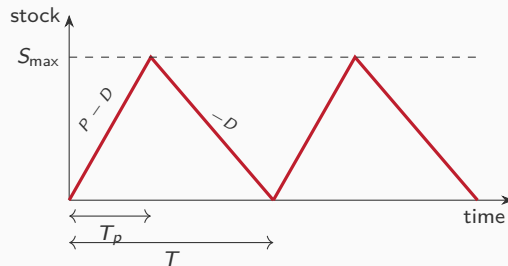
7.3 Inventory management: simultaneous production and consumption

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When we produce it ourselves

The factory produces its interior paint in **batches**. While a batch is being produced, sales continue: the stock rises at rate $P - D$ (production minus demand) and then falls at rate D .

- demand $D = 24\,000$ L/year;
- production capacity $P = 60\,000$ L/year;
- set-up of each batch (cleaning, colour adjustment): $C_e = 900$ €;
- production cost 2 €/L; holding 25%/year: $C_p = 0.50$ €/L/year.



$T_p = Q/P$: production time; $T = Q/D$: cycle; the stock never reaches Q , because part of it is sold while it is being produced.

The production and consumption model

Formulas

Maximum stock: $S_{\max} = Q \left(1 - \frac{D}{P}\right)$

$$K(Q) = DC_a + \frac{D}{Q} C_e + \frac{S_{\max}}{2} C_p$$

$$Q^* = \sqrt{\frac{2DC_e}{C_p}} \cdot \sqrt{\frac{P}{P-D}}$$

With $P \rightarrow \infty$ (the whole batch arrives at once) we are back to the EOQ of 7.1.

Why the optimal batch is larger: producing and selling at the same time lowers the average stock ($S_{\max}/2 < Q/2$), so each litre in a batch costs less to store — and it pays to space the set-ups further apart.

The factor $\sqrt{P/(P-D)}$ measures the effect: here $\sqrt{60/36} = 1.29$.

Production launch point: here we do not order, we launch a batch. If the set-up takes t_{prep} , we start setting it up when the stock falls to $D t_{\text{prep}}$.

Result: the interior paint

$Q^* = 9295 \times \sqrt{60/36}$	12 000 L
maximum stock $Q^*(1 - D/P)$	7200 L
production time $T_p = Q^*/P$	2.4 months
cycle $T = Q^*/D$	6 months
batches per year	2
set-up + holding	$1800 + 1800 = 3600$ €
total cost (with production)	51 600 €

Policy: two batches of 12 000 L per year; each one occupies the line for 2.4 months.

Comparison

If the whole batch arrived at once (EOQ of 7.1): $Q = 9295$ L and management cost 4648 €/year. Production simultaneous with consumption reduces it to 3600 € and increases the optimal batch by 29%.

Another reading: the line is only busy 40% of the time (D/P). In the other 60% it can produce other colours — planning several products on the same line is the next step (*economic lot scheduling*).

The chapter's four models side by side

Situation	Q^*	what changes
Purchase, no shortages (Wilson)	$\sqrt{2DC_e/C_p}$	—
Purchase, planned shortages	$\sqrt{2DC_e/C_p} / \sqrt{\rho}$, $\rho = \frac{C_r}{C_p+C_r}$	larger batch; P_e can be negative
Purchase with discounts	EOQ per price break, adjusted	the purchase enters the cost
Production and consumption	$\sqrt{2DC_e/C_p} \sqrt{P/(P-D)}$	larger batch; $S_{\max} < Q$

- In all of them, the cost is **flat** near the optimum (square root): moderate errors in the data cost little.
- All of them assume constant, known demand. With random demand a **safety stock** is needed — sized with the service-level ideas of ch. 6.

```
from math import sqrt

def producao(D, P, Ce, Cp, Ca=0):
    Q = sqrt(2 * D * Ce / Cp) * sqrt(P / (P - D))
    Smax = Q * (1 - D / P)
    K = D * Ca + D * Ce / Q + Cp * Smax / 2
    return Q, Smax, Q / D, Q / P, K    #  $Q$ ,  $S_{max}$ ,  $T$ ,  $T_p$ ,  $K$ 

print(producao(24000, 60000, 900, 0.5, 2))
# (12000.0, 7200.0, 0.5, 0.2, 51600.0)
```

Full file: code/python/07_inventory/
07_3_production/producao_consumo.py.

Try it: what happens to Q^* when P
approaches D ? Interpret.

Example 5 — synthetic resin

A company can produce 20 000 t of synthetic resin per year, whose average demand, over the same period, is 15 000 t. The set-up cost (5 days) of a batch is 98 000 m.u. and the production cost is 20 000 m.u./t. With a holding rate of 30% and a 300-day year, compute the optimal inventory management policy for this resin (optimal batch, cycle, production time, maximum stock, production launch point — the stock level at which the set-up of the batch should start — and annual cost).

Example 6 — the same model, other data

Production capacity 25 000 t/year; demand 20 000 t/year; set-up cost 76 000 m.u. (5 days); production cost 5000 m.u./t; holding rate 25%; 300-day year. Find the optimal policy.

KEY TAKEAWAYS — 7.3

1. Production and consumption: $S_{\max} = Q(1 - D/P)$; $Q^* = \sqrt{2DC_e/C_p} \sqrt{P/(P - D)}$; $T_p = Q/P$, $T = Q/D$; launch point = $D \times$ set-up time.
2. Producing while selling lowers the average stock and increases the optimal batch.
3. The four models have the same structure: ordering/set-up cost against holding cost (and shortage cost, or price).
4. Assumption common to all of them: constant, known demand.

End of chapter 7. Next: 8 Multi-criteria decision making (AHP).

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Inventory Theory (EOQ with gradual replenishment).
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017.
- Silver, E. A., Pyke, D. F., Thomas, D. J., *Inventory and Production Management in Supply Chains*, 4th ed., CRC Press, 2017.

Resources

- Exercise sheet 7. `code/python/07_inventory/07_3_production/producao_consumo.py`.